

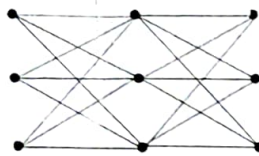
TEZPUR UNIVERSITY
Spring Semester End Exam. 2022
Graph Theory (CO 217)

Maximum Marks : 60

Duration : 3 hours

You may answer as many questions as you like but maximum you get is 60.

1. Draw a graph planar G which is 5-colorable but not 4-colorable. [4]
2. Suppose a forest has 1000 vertices and 800 edges. How many components does it have? [4]
3. $V - S$ is a independent set of a graph G iff S is a vertex cover G . [5]
4. Is the following graph Hamiltonian? Explain with property of Hamiltonicity. [5]

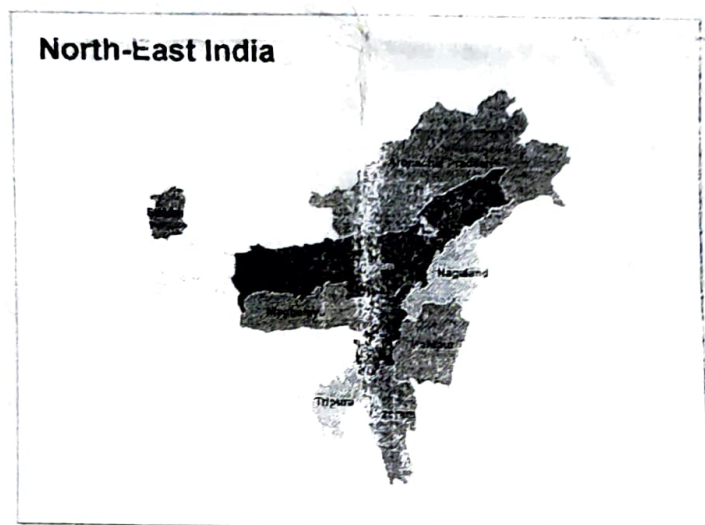


5. Consider the graph G below with 5 faces. If four edges are added to this graph so that the graph remains planar. What will be the maximum and minimum number of faces in resultant graph? Explain. [3+3]



6. State whether the following statements are true or false. You will get a ZERO if there are no proper explanations. [4 × 8]
 - i) If $\chi(G) = 2$, then the graph G is planar.
 - ii) The complement $\overline{C_n}$ of the n -cycle C_n is Hamiltonian when $n \geq 5$.
 - iii) If a graph G is 5-edge-connected, then G is 3-connected.
 - iv) If $\chi(G) > 2$, then the graph G must have a cycle.
 - v) A connected planar graph G has 7 faces; 3 of the faces have degree 4, 2 of the faces have degree 3 and 2 of the faces have degree 2. Number of edges in G is 11.
 - vi) If a graph G on n vertices has fewer than $2n - 4$ faces, then G is planar.
 - vii) If v and w are non-adjacent vertices of a Hamiltonian graph G with n vertices, then $\deg(v) + \deg(w) \geq n$.
 - viii) For any graph G , if $\delta(G) = 3$ then vertex connectivity of $\kappa(G) \geq 2$.

7. a) If G is a planar graph on $n \geq 11$ vertices, then its complement is not planar. [7]
 b) Prove that a 3-connected graph has a even cycle. [7]
 c) The join of two graphs $G + H$ is their graph union $G \cup H$ in addition to all the edges that connect each vertex of G with every vertex in H . Prove that $\chi(G + H) = \chi(G) + \chi(H)$. [5]
8. Answer the following questions related the map of the North-east India. [3+3+3+3]
 (a) Make a graph representing the map. (b) Is your graph 2-colorable? Explain. (c) Does your graph have an Euler Circuit? Explain. (d) What is the vertex connectivity of your graph?



Some Definitions

- The (vertex) connectivity number $\kappa(G)$ of a graph G is defined as:
 $\kappa(G) = \min\{k \mid k = |S|, G - S \text{ is disconnected or trivial}, S \in V(G)\}.$
- $\bar{G}$ is the complement of G if they have the same vertex set and an edge (u, v) is present in $\bar{G}$, if and only if the edge is not present in G . It means, two vertices are adjacent in $\bar{G}$ if the two vertices are not adjacent in G .
- The join of two graphs $G + H$ is:



- In a proper coloring of a graph two end points of each edge are colored with different colors. A graph G is k -colorable if G can be properly colored with k colors.

$$\chi(G) = \min\{k \mid G \text{ is } k\text{-colorable}\}$$